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### **1.0 - Executive Summary**

This project has the objective of simulating processes through Aspen Plus, and it was divided into two parts. The first part involves the use of “Refrigeration System with Flash Tank Economizer and Two Stages of Compression”, where the objective was to utilize the previous homework problem and transform it into a three-stage compression system. After carefully deciding on the new production design, the system’s interstage pressure must be optimized while removing 3MW of heat in the chiller by changing the flow rate of the refrigerant, followed by an analysis of COP (coefficient of performance). The reason of those changes is to make the system more efficient, where the power required by the compressors are lower in the three-stage compression system when compared to the two-stage design, also lowering the cost of operation and increasing the coefficient of performance. The new optimized interstage pressure is 442 kW with a COP of 2.18.

The last part was the Cyclohexane Production, where design specifications were created in order to meet the target ratios and a production of 300 KMTA final product by varying the Hydrogen to Benzene ratio and the recycle stream. This part also utilizes the problem of the previous homework, however the flow diagram had to be modified to show all the equipment involved in the design. In addition to those changes, the objective of this problem is to calculate the CAPEX (capital cost), OPEX (operation cost), including utilities and raw materials, revenues from the product and credits for fuel. In total, this plant will cost. The total cost of equipment was estimated as \$2,062,546, while the annual operating cost was of \$3,504,094 including utilities. More details about cost breaking down and credits towards the final product and fuel produced can be found in Appendix IIIB.

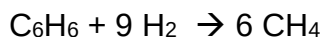
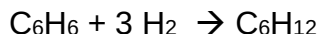
## 2.0 – System and Procedure

**Part A:** The new system includes a chiller, three compressors, two mixers, a condenser, two separators and three throttling valves. It utilizes propane as the working fluid throughout the system. More details can be found in the Process Flow Diagram in Appendix II-C. In order to remove the target of 3MW in the chiller, the optimal flow rate was calculated as follows:

1. The template was obtained from previous homework.
2. All the compressors were set to operate at 75% isentropic efficiency.
3. Pressure of S14 (outlet compressor 1 stream) and S5 (inlet mixer 1 stream) were assumed to be the same.
4. Pressure of S12 (outlet compressor 3 stream) and S8 (inlet mixer 2 stream) were assumed to be the same.
5. Pressure of S10 (outlet valve 3 stream) and S11 (chiller outlet stream) were assumed to be the same.
6. Tear stream was set as S2 (compressor 2 outlet stream).

**Part B:** The system includes two pumps, a compressor, a mixer, a bubble column reactor, a throttling valve, two splitters, a cooler, two heaters and two separation columns. More details can be found in the Process Flow Diagram in Appendix II-D.

In the original problem, benzene enters as a fresh feed at 100°F and 15 Psia by a supply stream (100 lbmol/h), followed by a secondary stream of fresh feed (313 lbmol/h) containing 97.5 % hydrogen gas (305.175 lbmol/h), 0.5% nitrogen gas (1.565 lbmol/h) and 2% methane gas (6.26 lbmol/h) at 120°F and 335 Psia. The two streams are combined with two recycling streams from the first high pressure separator, and all streams are mixed inside a mixer. The mix is then fed to the reactor where two reactions take place at 400°F and 315 Psia:



The resultant stream from the reactor is then cooled to 120°F and fed to the high-pressure separator at 315 Psia, where a vapor stream comes out of the top and a liquid stream comes out of the bottom. The top stream is followed by a purge and a recycle stream back to the mixer. The bottom stream is also split, where the resulting streams will be a liquid recycle back to the mixer and a secondary stream as the feed for the low-pressure separator. At the low-pressure separator, the final product will be collected out of the bottom stream, which may also include some impurities. The top vapor stream is composed mainly of methane.

The system presents some parameters that must be met in order to be considered a successful result. The vapor recycled stream must be adjusted in order to maintain a hydrogen to Benzene molar ratio of 3.3 in the mixer, while the liquid recycled stream must be no more than 45% of the fresh feed liquid Benzene flow rate. In addition to the original problem, the flowsheet must be reconfigured to show all the unit operations corresponding to the actual equipment in the field, as well as increasing the production of cyclohexane to 300 KMTA, while meeting all the requirements described above. A list with all the utilities and total operation cost for this product is also required, which includes the cost of raw material, credits for fuel production and revenue from the product.

For this assignment, the procedure was as follows:

1. The process flow diagram was obtained from a previous assignment.
2. Flowsheet was modified to include all the unit operations needed.
3. For the first design specification the purge stream should change in order to keep a molar ratio of 3:3 hydrogen to benzene, based on the tear stream. Since the tear stream composition is the same as the composition inside the mixer.
4. The second design specification was created around the liquid recycle, where the recycle stream was set as 45% of the fresh liquid benzene flow rate.
5. A calculator was set to calculate the fresh feed going inside the reactor based on the amount of product desired.
6. Two heaters were set as a heater exchanger.
7. A water cooler was set up in the system.
8. Temperature of the cooling water into the cooler was calculated.
9. New set of units was created to output the flow rate as KMTA.
10. All the costs were calculated.

### 3.0 – Physical Properties

| Property Name             | Units                     | C3H8       |
|---------------------------|---------------------------|------------|
| Molecular Weight          |                           | 44.0970    |
| Critical Temperature      | C                         | 96.6500    |
| Critical Pressure         | atm                       | 41.9000    |
| Critical Volume           | cm <sup>3</sup> /mol      | 203.4945   |
| Compressibility Factor    |                           | 0.2810     |
| Accentric Factor          |                           | 0.1520     |
| Normal Boiling Point      | C                         | -42.0500   |
| Freezing Point            | C                         | -187.6500  |
| Liquid Molar Volume       | cm <sup>3</sup> /mol      | 75.7832    |
| Enthalpy of Formation     | kcal/mol                  | -24.8200   |
| Gibbs Energy of Formation | kcal/mol                  | -5.6100    |
| Heat of Vaporization      | kcal/mol                  | 4.4870     |
| Heat Capacity             | cal/(mol C)               |            |
| A1                        |                           | -1.0090    |
| A2 x 1.0E3                |                           | 73.1500    |
| A3 x 1.0E6                |                           | -37.8900   |
| A4 x 1.0E9                |                           | 7.6780     |
| A5 x 1.0E12               |                           | 0.0000     |
| A6 x 1.0E15               |                           | 0.0000     |
| Low Temperature Limit     | C                         | 26.8500    |
| Upper Temperature Limit   | C                         | 1371.8500  |
| Liquid Vapor Pressure     | Pa                        |            |
| A1                        |                           | 63.5869    |
| A2                        |                           | -3550.1941 |
| A3                        |                           | 0.0000     |
| A4                        |                           | 0.0082     |
| A5                        |                           | -7.0923    |
| A6                        |                           | 0.0000     |
| A7                        |                           | 6.0000     |
| Low Temperature Limit     | C                         | -83.6707   |
| Upper Temperature Limit   | C                         | 96.6500    |
| Transport Properties      |                           |            |
| Diffusion Coefficient     | m <sup>2</sup> /s         |            |
| Mass Transfer Coefficient | m/s                       |            |
| Heat Transfer Coefficient | kcal/(m <sup>2</sup> s C) |            |
| Viscosity                 | cp                        |            |
| Thermal Conductivity      | kcal/(m s C)              |            |
| Solid State               |                           | 0.0        |

**Figure 1:** Physical Properties of Propane (Problem A)

| Property Name             | Units                     | H2        | N2        | CH4        | C6H6       | C6H12_1    | H2O        |
|---------------------------|---------------------------|-----------|-----------|------------|------------|------------|------------|
| Molecular Weight          |                           | 2.0160    | 28.0130   | 16.0430    | 78.1140    | 84.1620    | 18.0150    |
| Critical Temperature      | C                         | -239.9500 | -146.9500 | -82.5500   | 288.9500   | 280.2500   | 374.1500   |
| Critical Pressure         | atm                       | 12.8000   | 33.5000   | 45.4000    | 48.3000    | 40.2000    | 217.6000   |
| Critical Volume           | cm <sup>3</sup> /mol      | 64.9115   | 89.6408   | 99.2094    | 258.7791   | 308.3679   | 55.8953    |
| Compressibility Factor    |                           | 0.3050    | 0.2900    | 0.2880     | 0.2710     | 0.2730     | 0.2290     |
| Accentric Factor          |                           | -0.2200   | 0.0400    | 0.0080     | 0.2120     | 0.2130     | 0.3440     |
| Normal Boiling Point      | C                         | -252.7500 | -195.7500 | -161.4500  | 80.1500    | 80.7500    | 100.0000   |
| Freezing Point            | C                         | -259.1500 | -209.8500 | -182.4500  | 5.5500     | 6.5500     | 0.0000     |
| Liquid Molar Volume       | cm <sup>3</sup> /mol      | 28.6127   | 34.7065   | 37.7482    | 95.5084    | 116.6608   | 19.6361    |
| Enthalpy of Formation     | kcal/mol                  | 0.0000    | 0.0000    | -17.8900   | 19.8200    | -29.4300   | -57.8000   |
| Gibbs Energy of Formation | kcal/mol                  | 0.0000    | 0.0000    | -12.1500   | 30.9900    | 7.5900     | -54.6400   |
| Heat of Vaporization      | kcal/mol                  | 0.2160    | 1.3330    | 1.9550     | 7.3520     | 7.1600     | 9.7170     |
| Heat Capacity             | cal/(mol C)               |           |           |            |            |            |            |
| A1                        |                           | 6.1038    | 7.4400    | 4.5980     | -8.1010    | -13.0270   | 8.0582     |
| A2 x 1.0E3                |                           | 3.5399    | -3.2400   | 12.4500    | 113.3000   | 146.0000   | -1.6761    |
| A3 x 1.0E6                |                           | -5.3789   | 6.4000    | 2.8600     | -72.0600   | -60.2700   | 6.5196     |
| A4 x 1.0E9                |                           | 4.2220    | -2.7900   | -2.7030    | 17.0300    | 3.1560     | -3.9760    |
| A5 x 1.0E12               |                           | -1.4176   | 0.0000    | 0.0000     | 0.0000     | 0.0000     | 1.0265     |
| A6 x 1.0E15               |                           | 0.1704    | 0.0000    | 0.0000     | 0.0000     | 0.0000     | -0.0996    |
| Low Temperature Limit     | C                         | -73.1500  | 26.8500   | 26.8500    | 26.8500    | 26.8500    | -73.1500   |
| Upper Temperature Limit   | C                         | 2726.8500 | 936.0500  | 1367.8500  | 1137.3500  | 1082.3500  | 2726.8500  |
| Liquid Vapor Pressure     | Pa                        |           |           |            |            |            |            |
| A1                        |                           | 25.3527   | 46.1539   | 43.3183    | 73.8624    | 73.3379    | 65.1544    |
| A2                        |                           | -142.8939 | -984.2006 | -1380.5780 | -5970.4390 | -5885.9268 | -6842.9072 |
| A3                        |                           | 0.0000    | 0.0000    | 0.0000     | 0.0000     | 0.0000     | 0.0000     |
| A4                        |                           | 0.0971    | 0.0232    | 0.0116     | 0.0055     | 0.0055     | 0.0028     |
| A5                        |                           | -2.9194   | -5.4530   | -4.3964    | -8.0798    | -8.0304    | -6.1364    |
| A6                        |                           | 0.0000    | 0.0000    | 0.0000     | 0.0000     | 0.0000     | 0.0000     |
| A7                        |                           | 6.0000    | 6.0000    | 6.0000     | 6.0000     | 6.0000     | 6.0000     |
| Low Temperature Limit     | C                         | -258.6136 | -209.8500 | -182.4500  | 20.2109    | 19.5774    | 46.1166    |
| Upper Temperature Limit   | C                         | 2726.8500 | 936.0500  | 1367.8500  | 1137.3500  | 1082.3500  | 2726.8500  |
| Liquid Vapor Pressure     | Pa                        |           |           |            |            |            |            |
| A1                        |                           | 25.3527   | 46.1539   | 43.3183    | 73.8624    | 73.3379    | 65.1544    |
| A2                        |                           | -142.8939 | -984.2006 | -1380.5780 | -5970.4390 | -5885.9268 | -6842.9072 |
| A3                        |                           | 0.0000    | 0.0000    | 0.0000     | 0.0000     | 0.0000     | 0.0000     |
| A4                        |                           | 0.0971    | 0.0232    | 0.0116     | 0.0055     | 0.0055     | 0.0028     |
| A5                        |                           | -2.9194   | -5.4530   | -4.3964    | -8.0798    | -8.0304    | -6.1364    |
| A6                        |                           | 0.0000    | 0.0000    | 0.0000     | 0.0000     | 0.0000     | 0.0000     |
| A7                        |                           | 6.0000    | 6.0000    | 6.0000     | 6.0000     | 6.0000     | 6.0000     |
| Low Temperature Limit     | C                         | -258.6136 | -209.8500 | -182.4500  | 20.2109    | 19.5774    | 46.1166    |
| Upper Temperature Limit   | C                         | -239.9500 | -146.9500 | -82.5500   | 288.9500   | 280.2500   | 374.1500   |
| Transport Properties      |                           |           |           |            |            |            |            |
| Diffusion Coefficient     | m <sup>2</sup> /s         |           |           |            |            |            |            |
| Mass Transfer Coefficient | m/s                       |           |           |            |            |            |            |
| Heat Transfer Coefficient | kcal/(m <sup>2</sup> s C) |           |           |            |            |            |            |
| Viscosity                 | cp                        |           |           |            |            |            |            |
| Thermal Conductivity      | kcal/(m s C)              |           |           |            |            |            |            |
| Solid State               |                           | 0.0       | 0.0       | 0.0        | 0.0        | 0.0        | 0.0        |

**Figure 2:** Physical Properties of Hydrogen, Nitrene, Methane, Benzene, Cyclohexane and Water (Problem B).

## 4.0 – Components

For each process the followed components were involved.

### Part A:

- $C_3H_8$  – propane

### Part B:

- $H_2$  – Hydrogen gas
- $N_2$  – Nitrogen gas
- $CH_4$  – Methane
- $C_6H_6$  – Benzene
- $C_6H_{12}$  – Cyclohexane
- $H_2O$  – Water

## 5.0 – Equipment

### Part A:

For the Refrigeration with Flash Tank Economizer and Three Stages of Compression the following equipment were used:

- Three compressors. Compressor 1 changed the pressure of the propane to 7.41 bar before the mixer 1 and after the chiller, equipment has a power of 442 kW. Compressor 2 changed the pressure of the propane to 12.21 bar after the chiller and before the condenser, equipment has a power of 406 kW. Compressor 3 changed the pressure of the propane to 8.93 bar after the chiller and before the mixer 2, equipment has a power of 524 kW.
- Condenser: cooled down the propane stream from 59.64°C to 35 °C, equipment has duty of 4372 kJ/sec
- Throttling valves: can be used to isolate the stream and stop the process, it also is responsible for a small pressure drop. System currently has three of those valves. Valve 1 changed the pressure of the propane from 12.21 bar to 7.41 bar. Valve 2 changed the pressure of the propane from 7.41 bar to 3.81 bar. Valve 3 changed the pressure of the propane from 3.81 bar to 1.38 bar.
- Separators: vessel where the separation of the liquid and vapor propane occur, both separators have a duty of 0kJ/sec
- Chiller: Converted the liquid propane into gas, equipment has a duty of 3000 kJ/sec as specified by the problem.

- Mixer: place where two different streams meet and are combined into one.

**Part B:**

- Two pumps: first pump is used in the fresh benzene stream, where its pressure goes from 15 Psia to 335 Psia. Second pump is used in the liquid benzene recycle stream, where its pressure goes from 310 Psia to 335 Psia.
- Compressor: placed in the recycling gas stream, pressure goes from 310 Psia to 335 Psia.
- Mixer: Mix all the four streams into one, fresh feed hydrogen gas, fresh feed benzene and two recycling streams.
- Heater exchanger: this equipment was designed as two heaters, where the second one (hot side) receives the heat from the top one (cold side), symbolized by Stream 11. There is also a 5 Psia pressure drop inside the heater.
- Reactor: vessel where the chemical reactions took place. Chemical reactions occurred at 400°F and 315 Psia.
- Cooler: Utilizes cold water from a cooling tower nearby. It cools the stream from 157.42°F down to 120°F.
- High pressure separator followed by splitters: vessel where the first separation took place. Separation occurred at 120°F and 310 psia, with an overhead gas purge fraction of 0.1391 and a bottom liquid recycle fraction of 0.3035.
- Low pressure separator: where the final separation took place, and the final product was collected. Separation occurred at 15 psia with a product flow rate of 300 KMTA.



## 6.0 – Discussion and Analysis

### Part A:

For this problem the flow sheet diagram was downloaded from previous assignment and used in Aspen Plus, where three design specifications and two transfers were developed. After selecting the tear stream as stream 2, the design specs set the stream 10 to have the same pressure as stream 11. The second specification set the duty of the chiller to 3MW by varying the flow rate, while the third one set the stream 7 to have the same pressure as stream 12.

One of the transfers set the pressure of the compressor 1 equal to the output pressure of valve 1, while the other transfer set the pressure of compressor 2 equals to the pressure of stream 3. In this part was possible to use transfer since both valve 1 and stream 3 occurs before their designed comparison due to the process of this system. The same can't be said about stream 10 and 11 as well as stream 7 and 12, which determined the use of the system specification instead.

The coefficient of performance (COP) was calculated using the calculator function and printed out in the main sheet and later it was compared to the value found when only two compressors were used for the previous assignment. It is possible to verify that the coefficient of performance is higher for the system with three compressors than for the system with only two. This is due to the fact that setting up a second separator and recovering more vapor before sending the propane to the chiller, reduced the total power needed in the compressors. By adding a system in parallel to the one previously installed, it will help the performance of the plant. Formulas used can be found in Appendix IA and results can be found in Appendix IIIA.

### Part B:

For this problem, the flow sheet diagram was downloaded from previous homework and used in Aspen Plus. Modifications were made into the design to add two pumps, a compressor, a mixer, two heaters and a cooler. Two design specifications were created in order to obtain the production desired of 300 KMTA of a high purity product, and the stream going into the reactor was chosen to be the tear stream.

The first specification was created to set the recycle gas stream and the purge stream ratio. As a requirement from the system, the hydrogen rich recycle gas stream had to make up the concentration of the hydrogen in order to always keep a ratio of 3.3

hydrogen to benzene concentration on tear stream. By doing that, the purge stream was also created to discharge the excess hydrogen and other impurities in the system.

The second design specification was set as the liquid recycle been 45% of the fresh liquid benzene flow rate. A calculator was created to determine the raw material flow rate of the hydrogen and benzene stream needed in order to produce 300KMTA of cyclohexane.

The output pressure of both the pumps and the compressor were set in order to match the pressure of the fresh hydrogen feed going into the mixer. The resultant stream was sent into the heater, where was considered a pressure drop of 5 Psia since the reactor had an operation pressure set. The utility stream was added connecting both the heaters together, where the first heater was called cold side and the second one was called hot side.

The resultant stream of the heaters was then sent to the cooler where cold water was used to reduce the temperature of the stream. As the problem states, a cooling tower nearby was designed for a dry bulb temperature of 85°F with a relative humidity of 70% and an approach temperature of 10°C. A psychrometric chart [2] was used to determine the wet bulb temperature of the water as 72.2°F. It was also determined that the water could heat up to 122.2°F due to the approach temperature requirement. In order to set up the utility stream for the cooler, the utility tool from Aspen was used, all the information about the water stream was entered into the system, including the price of 0.5 \$/tonne [6].

To finalize this problem, a personalized set of units was created in Aspen, so the chart displaying the results would have the KMTA units for the flow rate.

## 7.0 – Appendix

### 7.1 – Appendix I: Formulas and Chart

#### 7.1.1 – Appendix IA

- Coefficient of Performance

$$COP = \frac{Q_{calc}}{W_1 + W_2 + W_3}$$

Where,

$Q_{calc}$ : duty calculated from the chiller

$W_1$ ,  $W_2$  and  $W_3$  power of compressor 1, compressor 2 and compressor 3 respectively.

- Ideal Coefficient of Performance

$$COP_{ideal} = \frac{T_{chiller}}{T_{cond} - T_{chiller}}$$

Where,

$T_{chiller}$ : temperature of the chiller in Kelvin

$T_{cond}$ : temperature of the condenser in Kelvin

#### 7.1.2 – Appendix IB

Since this problem is the only problem involving chemical reactions, the following were used for mass balance of the compounds

$$\text{Hydrogen: } n_{H_2} = n_{H_2}^0 - 3 * \mathfrak{J}_1 - 9 * \mathfrak{J}_2$$

$$\text{Nitrogen: } n_{N_2} = n_{N_2}^0$$

$$\text{Methane: } n_{CH_4} = n_{CH_4}^0 + 6 * \mathfrak{J}_2$$

$$\text{Benzene: } n_{C_6H_6} = n_{C_6H_6}^0 - 1 * \mathfrak{J}_1 - 1 * \mathfrak{J}_2$$

$$\text{Cyclohexane: } n_{C_6H_{12}} = n_{C_6H_{12}}^0 + 1 * \mathfrak{J}_1$$

- Conversion of Benzene

$$\beta_1 + \beta_2 = n_{\text{C}_6\text{H}_6}^0 * \text{conversion}$$

Where  $\beta_1$  and  $\beta_2$  are coefficients of reaction 1 and reaction 2 respectively.

Cost Analysis:

- Compressor

Equipment cost

$$\$ (2010) = 580,000 + 20,000 * (W(\text{kW}))^{0.6}$$

$$\$ (2020) = 1.4 * \text{Equip. Cost in 2010}$$

Installation Cost

$$\$ (2020) = 2.5 * \text{Equip. Cost in 2020}$$

Annual Cost

$$\$ (2020) = \frac{4}{3} * \text{installation cost in 2020}$$

Where,

$W(\text{kW})$  is the power of the compressor in kW.

- Heaters / Cooler

$$\Delta T_{\ln} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$A = \frac{Q}{Ft * U * \Delta T_{\ln}}$$

Equipment cost

$$\$ (2010) = 28,000 + 54 * (A(\text{m}^2))^{1.2}$$

$$\$ (2020) = 1.4 * \text{Equip. Cost in 2010}$$

Installation Cost

$$\$ (2020) = 2.5 * \text{Equip. Cost in 2020}$$

Annual Cost

$$\$ (2020) = \frac{4}{3} * \text{installation cost in 2020}$$

Where,

A is the heat transfer area in m<sup>2</sup>

Q is the duty of the heater

F<sub>t</sub> is 0.8 factor

U is the heat transfer coefficient

- Pump

Equipment cost

$$\$ (2010) = 8,000 + 240 * (Q(\frac{l}{s}))^{0.9}$$

$$\$ (2020) = 1.4 * \text{Equip. Cost in 2010}$$

Installation Cost

$$\$ (2020) = 4.0 * \text{Equip. Cost in 2020}$$

Annual Cost

$$\$ (2020) = \frac{4}{3} * \text{installation cost in 2020}$$

Where,

$Q(\frac{l}{s})$  is the flow rate in liters per second.

- Vertical Vessels

Assuming 5 minutes of residence time

Equipment cost

$$\$ (2010) = 11,600 + 34 * (M(Kg))^{0.85}$$

$$\$ (2020) = 1.4 * \text{Equip. Cost in 2010}$$

Installation Cost

$$\$ (2020) = 4.0 * \text{Equip. Cost in 2020}$$

Annual Cost

$$\$ (2020) = \frac{4}{3} * \text{installation cost in 2020}$$

Operating Pressure

$$\text{Thickness} = \frac{P_i * D_i}{2SE - 1.2 * P_i}$$

$$\text{Mass} = \pi * D_i * L * \text{thickness} * \rho$$

Volume of the vessel

$$V_{\text{vessel}} = \frac{\pi * D_i^3}{2}$$

Where,

M(Kg) is the total mass of the vessel in Kg.

P<sub>i</sub> is the internal operating pressure in Psig.

D<sub>i</sub> is the internal diameter.

E is the welded efficiency = 1

S is the stress of the vessel =  $88.9455 \frac{N}{mm^2}$

L is the length of the vessel.

P is the density of the material, stainless steel is  $8000 \frac{Kg}{m^3}$

- Utility Cost

Electricity

To calculate the total cost of electricity, the power of the three compressors were combined and converted to kw.

$$W_{\text{total}} = W_1 + W_2 + W_3$$

$$87 \text{ hp} * \frac{0.745 \text{ kw}}{1 \text{ hp}} = 64.8759 \text{ kw}$$

It was found that the average cost of electricity for industrial processes cost \$0.0657 / kw\*h [7]. It is estimate that the plant operates 24h a day, 330 days a year.

Cold water

To calculate the total cost of water needed for the cooler, the formula below was used:

$$Q = m * C_p * \Delta T$$

Where,

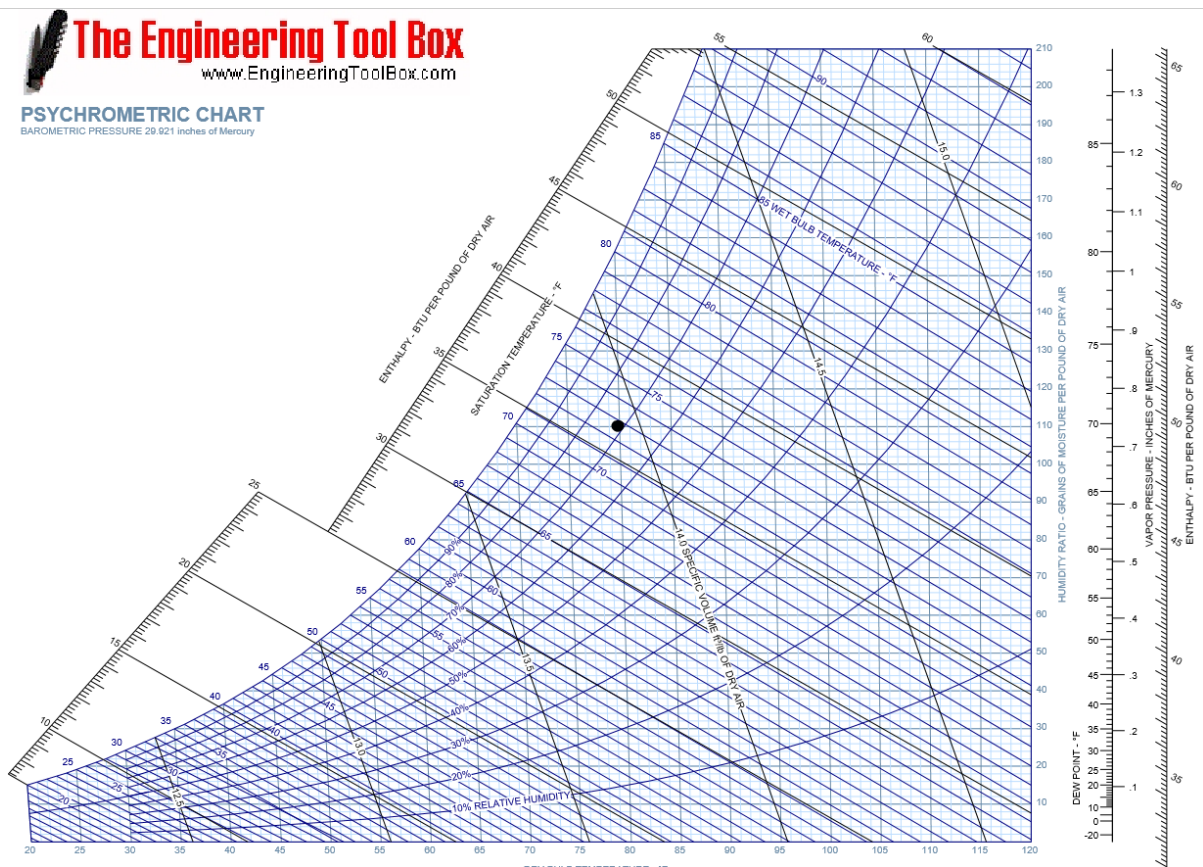
$Q$  is the heat removed in the cooler; the value was calculated by Aspen simulation.

$m$  is the mass flow rate of the water needed.

$C_p$  is the heat capacity of the water [3].

$\Delta T$  is the temperature variance of the water out – water in.

It was found that the average cost of water for industrial processes cost \$0.50 per tonne of water. It is estimate that the plant operates 24h a day, 330 days a year.



**Figure 3 [2]:** Psychrometric Chart for finding the wet bulb temperature of water





## 7.3 – Appendix III: Results

### 7.3.1 – Appendix IIIA

| Main Flowsheet × S1 (MATERIAL) - Stream Results (Custom) × + |           |       |            |  |
|--------------------------------------------------------------|-----------|-------|------------|--|
| Default                                                      |           |       |            |  |
|                                                              |           | Units | S1         |  |
| ▶ From                                                       |           |       | MIX1       |  |
| ▶ To                                                         |           |       | COMPR2     |  |
| ▶ Substream: MIXED                                           |           |       |            |  |
| ▶ Phase:                                                     |           |       | Vapor      |  |
| ▶ Component Mole Flow                                        |           |       |            |  |
| ▶ PROPANE                                                    | KMOL/HR   |       | 970.989    |  |
| ▶ Mole Flow                                                  | KMOL/HR   |       | 970.989    |  |
| ▶ Mass Flow                                                  | KG/HR     |       | 42817.3    |  |
| ▶ Volume Flow                                                | CUM/HR    |       | 2946.77    |  |
| ▶ Temperature                                                | C         |       | 34.4754    |  |
| ▶ Pressure                                                   | BAR       |       | 7.40592    |  |
| ▶ Vapor Fraction                                             |           |       | 1          |  |
| ▶ Liquid Fraction                                            |           |       | 0          |  |
| ▶ Solid Fraction                                             |           |       | 0          |  |
| ▶ Molar Enthalpy                                             | KJ/KMOL   |       | -104840    |  |
| ▶ Mass Enthalpy                                              | KJ/KG     |       | -2377.5    |  |
| ▶ Enthalpy Flow                                              | KJ/SEC    |       | -28277.3   |  |
| ▶ Molar Entropy                                              | KJ/KMOL-K |       | -285.351   |  |
| ▶ Mass Entropy                                               | KJ/KG-K   |       | -6.47105   |  |
| ▶ Molar Density                                              | MOL/CC    |       | 0.00032951 |  |
| ▶ Mass Density                                               | GM/CC     |       | 0.0145302  |  |

**Figure 6:** Propane Stream

| Main Flowsheet × Custom Tables - C-1 × + |          |                            |               |
|------------------------------------------|----------|----------------------------|---------------|
| Place table on flowsheet                 |          | Place as icon on flowsheet | Send to Excel |
| Tab 1                                    |          |                            |               |
|                                          | Name     | Units                      | Value         |
| ▶                                        | COP      |                            | 2.18736       |
| ▶                                        | COPOLD   |                            | 2.09059       |
| ▶                                        | COPIDEAL |                            | 3.40214       |

**Figure 7:** Results of the coefficients of performance.

## 7.3.2 – Appendix IIIB

| Material                                                                                        | Heat                      | Load              | Work              | Vol.% Curves     | Wt. % Curves       | Petroleum       | Polymers | Solids |
|-------------------------------------------------------------------------------------------------|---------------------------|-------------------|-------------------|------------------|--------------------|-----------------|----------|--------|
| Display: <div>Streams</div> Format: <div>FULL</div> <div>Stream Table</div> <div>Copy All</div> |                           |                   |                   |                  |                    |                 |          |        |
|                                                                                                 |                           | <div>BZFEED</div> | <div>H2FEED</div> | <div>PURGE</div> | <div>PRODUCT</div> | <div>FUEL</div> |          |        |
| ▶                                                                                               | Mole Flow lbmol/hr        |                   |                   |                  |                    |                 |          |        |
| ▶                                                                                               | H2                        | 0                 | 3041.09           | 25.3797          | 0.0190891          | 2.71873         |          |        |
| ▶                                                                                               | N2                        | 0                 | 15.5953           | 11.9913          | 0.0727243          | 3.53206         |          |        |
| ▶                                                                                               | C1                        | 0                 | 62.3812           | 44.8344          | 1.90595            | 33.6482         |          |        |
| ▶                                                                                               | BZ                        | 999.699           | 0                 | 0.00244504       | 1.35147            | 0.0247322       |          |        |
| ▶                                                                                               | CH                        | 0                 | 0                 | 1.67797          | 976.901            | 16.7408         |          |        |
| ▶                                                                                               | WATER                     | 0                 | 0                 | 0                | 0                  | 0               |          |        |
| ▶                                                                                               | Mole Frac                 |                   |                   |                  |                    |                 |          |        |
| ▶                                                                                               | H2                        | 0                 | 0.975             | 0.302551         | 1.94737e-05        | 0.0479794       |          |        |
| ▶                                                                                               | N2                        | 0                 | 0.005             | 0.142948         | 7.41896e-05        | 0.0623329       |          |        |
| ▶                                                                                               | C1                        | 0                 | 0.02              | 0.534469         | 0.00194435         | 0.593814        |          |        |
| ▶                                                                                               | BZ                        | 1                 | 0                 | 2.91472e-05      | 0.0013787          | 0.000436469     |          |        |
| ▶                                                                                               | CH                        | 0                 | 0                 | 0.0200029        | 0.996583           | 0.295437        |          |        |
| ▶                                                                                               | WATER                     | 0                 | 0                 | 0                | 0                  | 0               |          |        |
| ▶                                                                                               | Total Flow lbmol/hr       | 999.699           | 3119.06           | 83.8859          | 980.25             | 56.6645         |          |        |
| ▶                                                                                               | Total Flow ktonne/oper... | 3.15604e+06       | 3.15576e+06       | 3.15573e+06      | 3.15606e+06        | 3.15574e+06     |          |        |
| ▶                                                                                               | Total Flow cuft/hr        | 1455.86           | 58678.5           | 1676.03          | 1754.83            | 23117           |          |        |
| ▶                                                                                               | Temperature F             | 100               | 120               | 120              | 114.678            | 114.678         |          |        |
| ▶                                                                                               | Pressure psia             | 15                | 335               | 310              | 15                 | 15              |          |        |

**Figure 8:** Results of the streams for cyclohexane production.

|                  |             |  |  |
|------------------|-------------|--|--|
| Display: Streams |             |  |  |
| ▶                | S11         |  |  |
| ▶ QCALC Btu/hr   | -31947062.1 |  |  |
| ▶ TBEGIN F       | 106.769625  |  |  |
| ▶ TEND F         | 300.        |  |  |

**Figure 9:** Results from the heat exchange between the heaters.

| Cyclohexane Production Cost Estimate |                                                  |                            |                                              |                  |                |                    |
|--------------------------------------|--------------------------------------------------|----------------------------|----------------------------------------------|------------------|----------------|--------------------|
| EQUIPMENT LIST: CAPITAL COSTS        |                                                  |                            |                                              |                  |                |                    |
|                                      | <b>Project:</b> Assignment #4                    |                            | <b>Chemical Engineers:</b> Mariana St. Peter | Jasmine Villegas |                |                    |
|                                      | <b>Location:</b> Chemical Engineering Department |                            | <b>Date:</b> 10/22/2019                      |                  |                |                    |
| Equipment Class                      | ASPEN BlockID                                    | Service Description        | Tag                                          | Material Cost    | Installed Cost | Annual Cost, \$/yr |
| <b>Drums</b>                         |                                                  |                            |                                              |                  |                |                    |
| Vertical Tank                        | HPSEP                                            | High Pressure Separator    | V-1                                          | \$215,176        | \$860,706      | \$1,147,608        |
| Vertical Tank                        | LPSEP                                            | Low Pressure Separator     | V-2                                          | \$64,696         | \$258,784      | \$345,045          |
| <b>Heat Exchangers</b>               |                                                  |                            |                                              |                  |                |                    |
| S&T Exchanger                        | COLD-SID                                         |                            | E-1                                          | \$138,440        | \$484,540      | \$646,054          |
| S&T Exchanger                        | HOT-SIDE                                         |                            | E-2                                          | \$317,148        | \$1,110,020    | \$1,480,026        |
| <b>Water Coolers</b>                 |                                                  |                            |                                              |                  |                |                    |
| Water Cooler                         |                                                  | Cools Reactor Effluent     | EA-1                                         | \$273,172        | \$956,104      | \$1,274,805        |
|                                      |                                                  |                            | EA-2                                         |                  |                |                    |
| <b>Pumps</b>                         |                                                  |                            |                                              |                  |                |                    |
| Pump                                 | FBPUMP                                           | Fresh Benzene Pump         | P-1                                          | \$14,217         | \$56,869       | \$75,825           |
| Pump                                 | RBUMP                                            | Recycle Benzene Pump       | P-2                                          | \$12,946         | \$51,783       | \$69,045           |
| <b>Compressors</b>                   | H2COMPR                                          | Compress the Recycling Gas |                                              | \$1,026,750      | \$2,566,874    | \$3,422,498.91     |
| <b>Total</b>                         |                                                  |                            |                                              | \$2,062,546      | \$6,345,680    | \$8,460,906        |

Figure 10: Result for the capital costs of equipment for the cyclohexane production.

| Cyclohexane Production Cost Estimate |                                                  |                                         |                                              |                             |                         |                    |
|--------------------------------------|--------------------------------------------------|-----------------------------------------|----------------------------------------------|-----------------------------|-------------------------|--------------------|
| UTILITIES: MATERIAL COST             |                                                  |                                         |                                              |                             |                         |                    |
|                                      | <b>Project:</b> Assignment #4                    |                                         | <b>Chemical Engineers:</b> Mariana St. Peter | Jasmine Villegas            |                         |                    |
|                                      | <b>Location:</b> Chemical Engineering Department |                                         | <b>Date:</b> 10/22/2019                      |                             |                         |                    |
| Material Stream                      | ASPEN ID                                         | Service                                 | Tag                                          | Usage                       | Unit Cost               | Annual Cost, \$/yr |
| Hydrogen                             | H2FEED                                           | Hydrogen feed                           | HF-1                                         | 7568.4 lb/hr                | 0.46 \$/lb              | \$27,851,539       |
| Benzene                              | BZFEED                                           | Benzene feed                            | BF-1                                         | 78092.7 lb/hr               | 0.45 \$/lb              | \$281,133,612      |
| Cyclohexane                          | PRODUCT                                          | Cyclohexane product, 99.7% purity       | CH-1                                         | 82358.1 lb/hr               | 0.78 \$/lb              | \$513,914,606      |
| Methane                              | FUEL                                             | Fuel gas by-product, 60% methane        | FG-1                                         | 13870.6 ft <sup>3</sup> /hr | 6.23 \$/ft <sup>3</sup> | \$691,312,598      |
| UTILITIES: OPERATING COST            |                                                  |                                         |                                              |                             |                         |                    |
|                                      | <b>Project:</b> Assignment #4                    |                                         | <b>Chemical Engineers:</b> Mariana St. Peter | Jasmine Villegas            |                         |                    |
|                                      | <b>Location:</b> Chemical Engineering Department |                                         | <b>Date:</b> 10/22/2019                      |                             |                         |                    |
| Utility Class                        | ASPEN ID                                         | Service                                 | Tag                                          | Usage per hour              | Unit Cost               | Annual Cost, \$/yr |
| Cooling Water [kJ]                   | --                                               | Water used in the cooler                | CW-1                                         | 2381.43                     | \$0.5000                | \$9,430            |
| Power [kw]                           | --                                               | Power supplied to pumps and compressors | --                                           | 64.875                      | \$0.0657                | \$33,757           |
|                                      |                                                  |                                         |                                              |                             | Total Utility Cost:     | \$43,188           |
|                                      |                                                  |                                         |                                              |                             | Total Cost =            | \$314,980,514      |
|                                      |                                                  |                                         |                                              |                             | Total Revenue =         | \$1,205,227,204    |
|                                      |                                                  |                                         |                                              |                             | Total Profit =          | \$884,294,515      |

Figure 11: Results for the cost raw material and utilities in \$/year [1, 4, 5, 7]

## 7.0 – References

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